

KHARITA

Your map to higher education in Morocco

Project overview

What it is, who it is for, and why we built it

By Moroccan students, for Moroccan futures.

1. What is Kharita?

"Kharita" means **map**. Kharita is a free website that shows Moroccan students the roads they can take after their diploma: which schools exist, which programs they can enter, what each program leads to, and which jobs come at the end.

A student tells Kharita two things: the diploma they have today and the field they like. Kharita then draws every path from that diploma to the programs and jobs in that field, step by step.

2. Why we built it

This project started from a real story. One of our team members tells it:

"My dream was to work in cyber security. I loved computers and I wanted to understand how systems are attacked and protected. But nobody could tell me how to get there.

First I took a professional baccalaureate. I was told it was a good choice. It was not. It cost me time and energy and it did not bring me closer to my dream. After that I studied to become a specialised technician in civil engineering, a field that had nothing to do with what I really wanted.

I was lost. I did not know which school to choose, which diploma I needed, or which door was still open for me. I had to find everything alone, and I lost years doing it.

I know I am not the only one. Many Moroccan students feel the same: they have a dream, but no map. That is why we built Kharita. I always wanted to do something that has a real impact, and this is how we are trying to do it."

So the reason is simple: **no student should lose years because nobody showed them the way.**

3. Who is it for?

- **High school students** who must choose what to do after the baccalaureate.
- **Students who already have a diploma** (DUT, BTS, DEUG, licence...) and want to know what they can do next.
- **People who want to change direction** and need to see which doors are still open.
- **Parents and teachers** who want to give good advice.

Kharita is **free for everyone**. It works in **Arabic, French and English**.

4. What can you do with Kharita?

- **Find your path.** Choose your diploma and a field. See all the programs you can enter and where each one leads.
- **Choose a target job.** Pick the job you want and see only the paths that reach it.
- **See the details of each program.** Years of study, cost, entrance exam, age limit, minimum grade, and the jobs it leads to.
- **Explore schools on a map of Morocco.** See which universities and schools exist in each city.
- **Ask the chatbot.** Ask questions in your own words and get answers based on real career stories.
- **Create an account.** Save your diploma and your interests so Kharita shows your paths directly.
- **Contact the team.** Send us a message, a correction or an idea.

5. The parts of the project

Kharita was built by a team of five. Each person built one part. All parts share the same database and appear together in one website.

Part	Built by	What it does	Built with	Who it is for
Path finder and map	moel-oua	Draws the paths from a diploma to programs and jobs. Shows schools on a map of Morocco, with a page of details for each school.	Back: Python, Flask, Peewee. Front: React Flow, Leaflet.	Every student who wants to see their options.
Auth and accounts	Zakaria (zguellou)	Authentication: sign up, log in (also with Google), two-step verification and password reset. Accounts: the user profile, where each user saves their diploma and interests.	Back: Node.js, Express, JWT. Front: Next.js, React.	Users who want to save their diploma and interests safely.
Chatbot	tbenzaid	Answers questions using real stories of people and their careers.	Back: Python, FastAPI, ChromaDB, Sentence Transformers. Front: Next.js, React.	Students who prefer to ask a question instead of searching.
Home page, contact, and adding schools	yabounna	The home page. The contact form: sending a message (POST) and reading the messages (GET). The form to add a new university and a new program. Front and back for all of these.	Back: Python, FastAPI, SQLAlchemy. Front: Next.js, React.	New visitors, anyone who wants to write to us, and people who want to suggest a school or a program.
Admin dashboard and uploads	ctoujana	Lets admins check and approve universities, programs, diplomas, jobs and users. Stores the images.	Back: NestJS, Prisma, Sharp. Front: Next.js, React.	The team that keeps the information correct.
Database	The whole team	Keeps all universities, programs, diplomas and jobs in three languages.	PostgreSQL.	Every part of the website uses it.

Tools shared by everyone: the website is one Next.js and React application written in TypeScript, styled with Tailwind CSS, and translated with next-intl. Every part runs in its own Docker container, started together with Docker Compose.

6. How the parts work together

- A visitor opens the website and reads the **home page**.
- They can create an **account** and save their diploma and interests.
- The **path finder** reads that information and draws their paths.
- The **map** shows where the schools are.
- The **chatbot** answers the questions that are left.
- Admins use the **dashboard** to add new schools and programs, so the information stays up to date.

7. Where we are today

The website works from start to finish. Today it contains sample information: about 40 universities and schools in 14 cities, 100 programs and more than 40 jobs. This information was written by the team to test the website and must be checked against official sources before a public launch.

8. Our goal

We want every Moroccan student to open Kharita, enter their diploma, and see clearly where they can go. No cost, no account needed to look, and no need to know someone who knows.

We had a dream of making an impact. Kharita is how we are trying to achieve it.